Business Analytics Report

Executive Summary The business faces a critical juncture, defined by a debilitating 42.5% return rate, significant negative revenue growth of -11.65%, and low customer satisfaction (average 2.1). These issues severely undermine profitability and hinder growth. However, a strong 81.58% repeat customer base and a projected substantial increase in total customers to 205.43 present clear opportunities. Immediate and decisive action is required to address product quality, enhance customer satisfaction, and strategically leverage the existing loyal customer base to stabilize financials and capitalize on future growth prospects.

Source: AI-generated with Gemini

Descriptive Analytics - Total revenue: $16,602.56 - Total transactions: 200 - Total customers: 76 - Average order value: $83.01 - Monthly revenue growth: 41.6% - Repeat customer rate: 81.6% - Return rate: 42.5%

Predictive Analytics - Customer Segment Prediction: best available model is Random Forest Segment Classifier. - Return Prediction: best available model is Logistic Regression. - Satisfaction Prediction: best available model is Gradient Boosting Regressor with RMSE 0.64.

Sentiment Analysis Sentiment analysis covers 200 feedback records. Distribution: Positive: 73, Negative: 67, Neutral: 60.

Forecasting - Monthly Revenue Forecast: 793.72 next period using Prophet. - Transaction Forecast: 10.40 next period using Prophet. - Customer Growth Forecast: 205.43 next period using Prophet. - Satisfaction Forecast: 2.07 next period using Prophet.

Business Recommendations 1. Implement a comprehensive strategy to reduce the 42.5% return rate by investigating root causes related to product quality or description accuracy. 2. Launch an initiative to improve

the average customer satisfaction score from 2.1 by addressing key pain points identified through customer feedback. 3. Develop targeted engagement programs for the 81.58% repeat customer base to enhance loyalty and increase their average spend of $218.45.

Limitations - The analysis is based only on the supplied Rialto dataset. - Forecasts are constrained by the available monthly history. - AI-generated narratives are explanations of calculated analytics, not replacements for the calculations.

Future Improvements - Add more transaction history to strengthen forecasting accuracy. - Add controlled operational cost and margin data. - Monitor model performance after each new dataset refresh.